Generative AI for Healthcare

Causality in Medical Imaging

Dr Fabio S. Ribeiro

2/2/2026

Generative AI: Recent Progress

Prompt: A crab made of cheese on a plate

Stable Diffusion (Stability AI)

DALL·E (OpenAI)

Generative AI: Recent Progress

Prompt: An old rusted robot wearing pants and a jacket riding skis in a supermarket

Stable Diffusion (Stability AI)

DALL·E (OpenAI)

Generative AI: Recent Progress

Prompt: A whimsical and creative image depicting a hybrid creature that is a mix of a waffle and a hippopotamus…

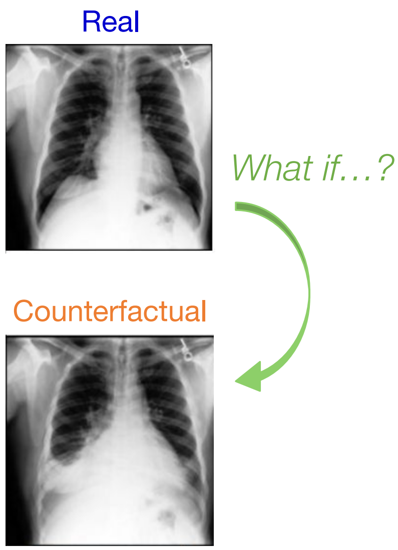
Stable Diffusion (Stability AI)

DALL·E (OpenAI)

Generative AI: For Medical Imaging!

“What would this medical scan have looked like if the patient had a disease?”

- ➔ **Understanding the data-generating process:** Move beyond correlation by answering “*what if?*” questions
- ➔ **Bias auditing:** Stress-testing AI models on different patient demographics (e.g. *age*, *race* etc)
- ➔ **AI interpretability:** Help explain which features drive decisions in AI systems
- ➔ **Synthetic data:** Improve AI models by augmenting datasets with synthetic examples



What is Causality?

- ➔ Scientific inquiry is often motivated by **causal** questions:

  I. How effective is a treatment in preventing a disease?

  II. If I take this pill, will my headache be gone?

  III. Would my grades have been better had I studied more?

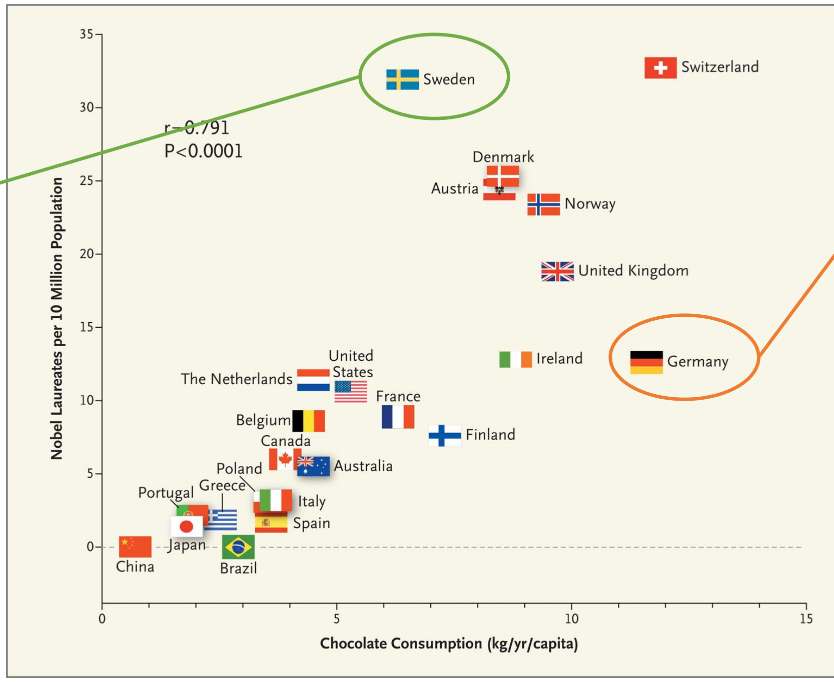
Causality is the relationship between cause and effect

Example: chocolate

"Correlation does *not* imply causation"

Chocolate Consumption vs Nobel Prize

Secretly the best chocolate?



Least effective chocolate?

Messerli. Chocolate Consumption, Cognitive Function, and Nobel Laureates. N Engl J Med 2012

Inspired by Jonas Peters

Reichenbach's Common Cause Principle

Hans Reichenbach

1. Chocolate consumption causes Nobel prize win

$$C \rightarrow N$$

2. Nobel prize win causes chocolate consumption

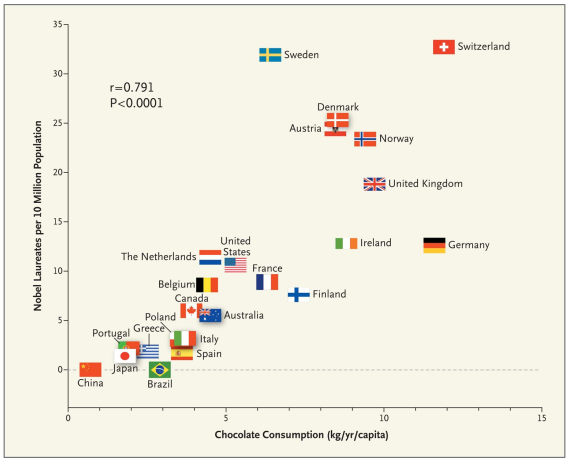
$$C \leftarrow N$$

3. Both are caused by an **unknown factor** U

$$C \leftarrow U \rightarrow N$$

Could U be GDP per capita or wealth per adult?

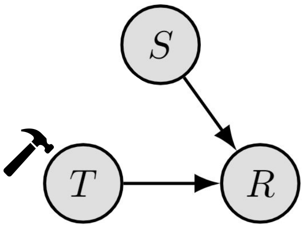
Chocolate Consumption vs Nobel Prize



Messerli. Chocolate Consumption, Cognitive Function, and Nobel Laureates. N Engl J Med 2012

Simpson's Paradox

Edward Simpson



- ➔ Stone size (S) is a **confounder**
- ➔ Make treatment (T) independent of size:
  - ◆ What's the recovery (R) rate if ***all*** subjects receive treatment **A** vs **B**?

Kidney Stone Size	Treatment A	Treatment B
Small	**93%** (81/87)	87% (234/270)
Large	**73%** (192/263)	69% (55/80)
Both	78% (273/350)	**83%** (289/350)

??

Patients with **large stones** received the **better treatment** (**A**), and those with **small stones** received the **inferior treatment** (**B**)

Charig CR, *et al*. Comparison of treatment of renal calculi by open surgery, percutaneous nephrolithotomy, and extracorporeal shockwave lithotripsy. Br Med J (Clin Res Ed). 1986 Mar 29.

Simpson's Paradox

Edward Simpson

Correlations may **reverse** depending on how we aggregate data and its subpopulations

From: https://upload.wikimedia.org/wikipedia/commons/f/fb/Simpsons_paradox_-_animation.gif

Predictive Modelling

Given an **image** X, train a **model** to predict some **label** Y

$$P(Y \mid X)$$

- ➔ Assumptions:
  - Sufficient training data (X, Y) is available
  - Training and test data come from the same distribution

Predictive Modelling: A Causal Perspective

What is the **causal relationship** between image X and label Y?

$$P(Y \mid X)$$

$$X \rightarrow Y$$

causal
(predict effect from cause)

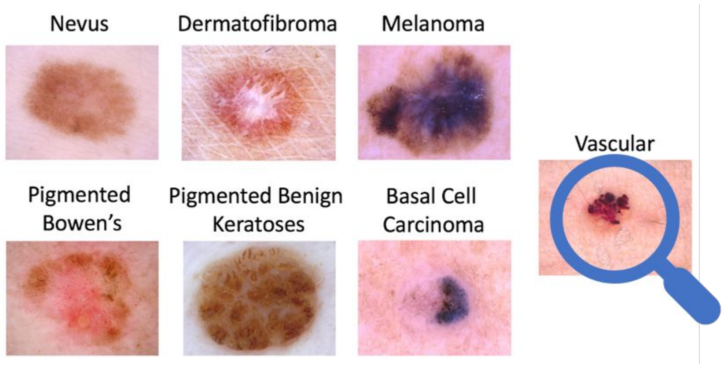
$$X \leftarrow Y$$

anti-causal
(predict cause from effect)

Example: Skin Lesion Classification

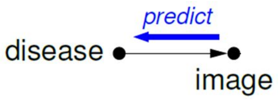
X – dermascopic image

Y – biopsy-derived diagnosis



$$X \leftarrow Y$$

anti-causal
(predict cause from effect)



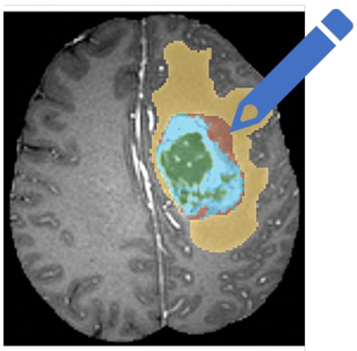
$$P(Y \mid X)$$

Image: ISIC 2018 challenge (https://challenge2018.isic-archive.com/task3/)

Example: Brain Tumour Segmentation

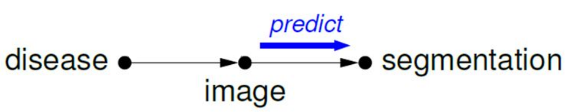
X – structural brain MRI

Y – manually drawn contour



$$X \rightarrow Y$$

causal
(predict effect from cause)



$$P(Y \mid X)$$

Image: ISIC 2018 challenge (https://challenge2018.isic-archive.com/task3/)

Example: Radiology Reports

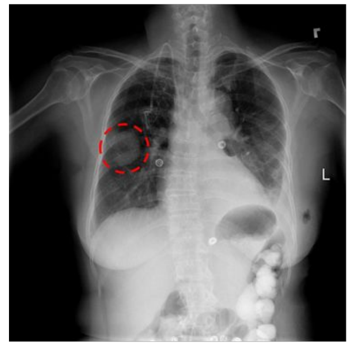
X – chest X-ray

Y – diagnosis extracted from report

$$X \overset{?}{\leftrightarrow} Y$$

causal or anti-causal?

$$P(Y \mid X)$$



Not clear whether the diagnosis was derived from the image or another lab result...

Image: ISIC 2018 challenge (https://challenge2018.isic-archive.com/task3/)

Example: Radiology Reports

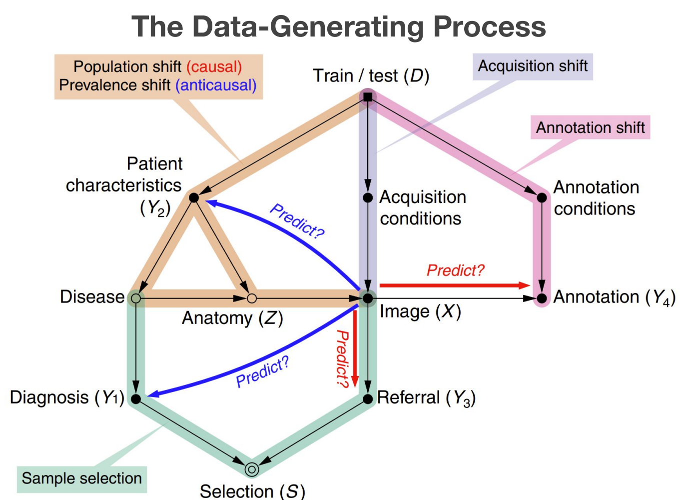
X – chest X-ray

Y – diagnosis extracted from report

$$X \overset{?}{\leftrightarrow} Y$$

causal or anti-causal?

$$P(Y \mid X)$$



Castro, Walker and Glocker. Causality matters in medical imaging. *Nature Communications* 2020

Meta information is needed to establish causal relationships, identify biases, and control for them

Ladder of Causation

Counterfactuals

What if X had happened?

I. Counterfactual reasoning
II. Deduce causes for observed events

3.

Intervention

What will Y be if i **do** X?

I. Use experiments to identify causal effects
II. Crucial for planning and policy making

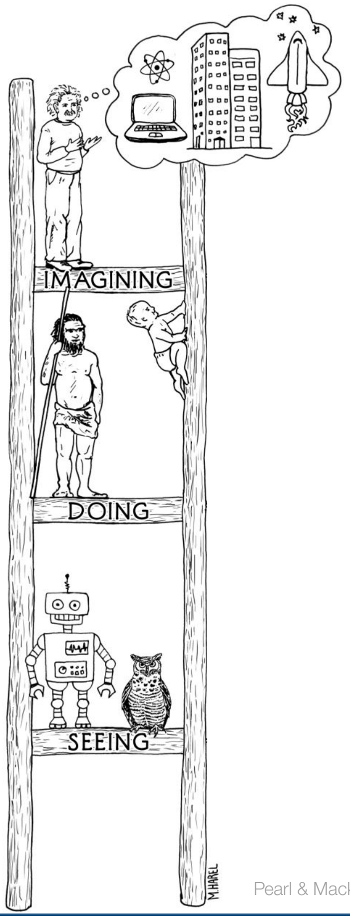
2.

Association

How would **seeing** X change my belief in Y?

I. Modelling correlations $P(Y|X)$
II. Modern AI excels at this task

1.



Judea Pearl

Pearl & Mackenzie. *The Book of Why*

Structural Causal Models

➔ A Structural Causal Model (SCM) is a triple: $\mathcal{M} := \langle X, U, F \rangle$

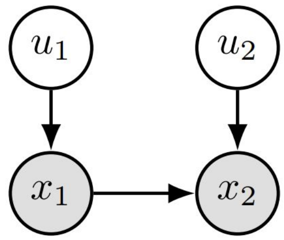
- Two sets of variables, observed and unobserved:

$$X = \{x_1, \ldots, x_N\} \qquad U = \{u_1, \ldots, u_N\}$$

Structural Causal Models

➔ Example: A simple SCM:

- x_1, x_2 are **endogenous** whereas u_1, u_2 are **exogenous**

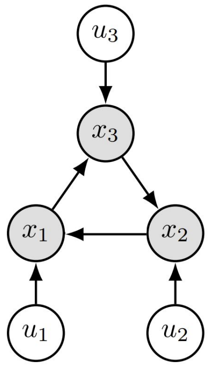


$$\boxed{x_1} := f_1(u_1), \qquad u_1 \sim \mathcal{N}(0,1)$$

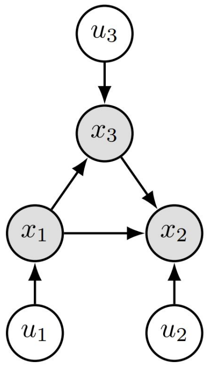
$$x_2 := f_2(\boxed{x_1}, u_2), \qquad u_2 \sim \mathcal{N}(0,1)$$

Structural Causal Models

➔ Acyclic SCMs can be represented by a **Directed Acyclic Graphs** (DAGs), with edges pointing from causes to effects



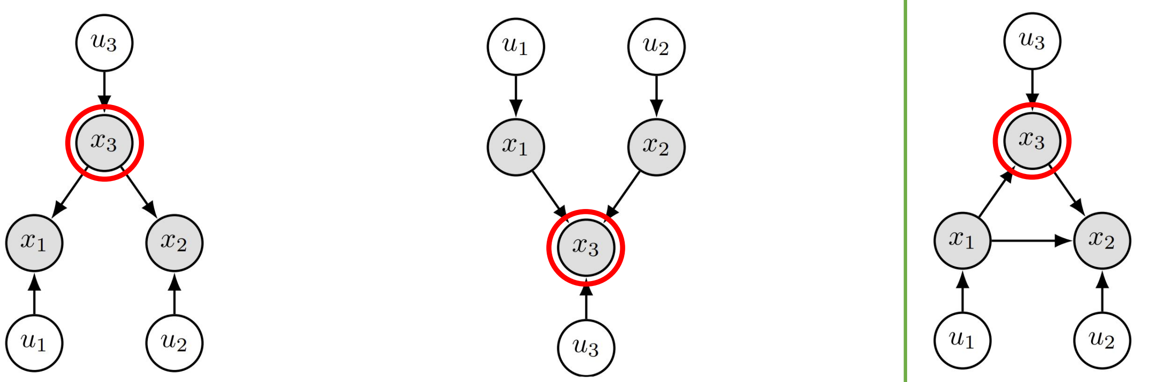
(a) Cyclic



(b) Acyclic

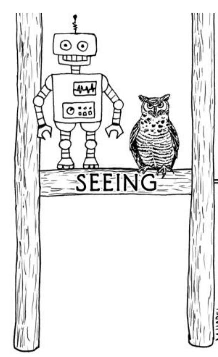
Structural Causal Models

➔ Acyclic SCMs can be represented by a **Directed Acyclic Graphs** (DAGs), with edges pointing from causes to effects



(a) Confounder (b) Collider (c) Mediator

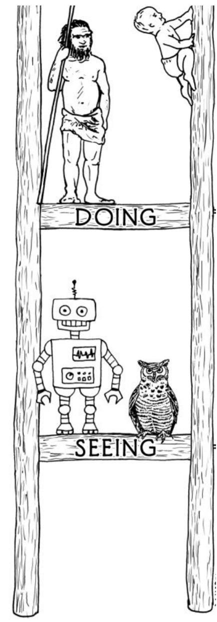
Observational Distribution (rung 1.)



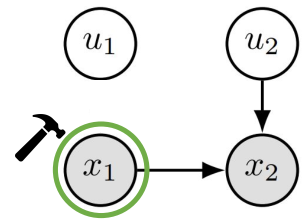
➔ SCMs with jointly **independent** exogenous noises are **Markovian**:

$$P(u_1, \ldots, u_N) = \prod_{k=1}^{N} P(u_k)$$

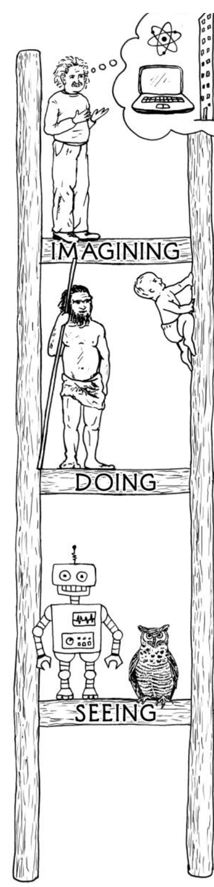
Interventional Distribution (rung 2.)



- ➔ SCMs **predict** the **causal effects** of actions via **interventions**
- ➔ **Interventions** answer causal questions like:
    - ◆ E.g. what would x_2 be if we set $x_1 := c$?

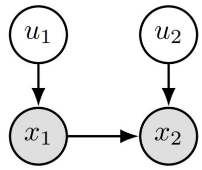


Counterfactuals (rung 3.)



- ➔ Causal models can consider hypothetical scenarios:
- - *“Arrived late, what if the train had been on time?”*
- ➔ Counterfactual inference involves three steps:
    1. **Abduction**: Infer why I was late (e.g. delayed train)
    2. **Action**: Imagine the train was on time
    3. **Prediction**: Predict what time I would have arrived

Example: Computing Counterfactuals



$$x_1 := f_1(u_1) = 1 + u_1$$

$$x_2 := f_2(x_1, u_2) = 3x_1 + u_2$$

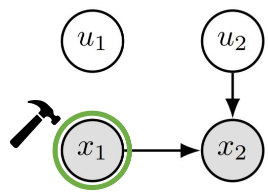
Q: Given we observed $\{x_1{=}2, x_2{=}4\}$, what would x_2 have been had x_1 been 5?

1. Abduction:

$$\boxed{x_1} := 2 = 1 + u_1 \implies u_1 = 1$$

$$x_2 := 4 = 3 \cdot \boxed{2} + u_2 \implies \boxed{u_2} = -2$$

2. Action: $\boxed{\widetilde{x}_1} := 5$



3. Prediction: $\widetilde{x}_2 = 3\boxed{\widetilde{x}_1} + \boxed{u_2} = 3 \cdot 5 - 2 = 13$

Generative AI for Causal Modelling

➔ Use generative AI to learn causal models

➔ Estimate counterfactuals, i.e. answer *"what if?"* questions

➔ Challenging in complex problems, e.g. medical imaging

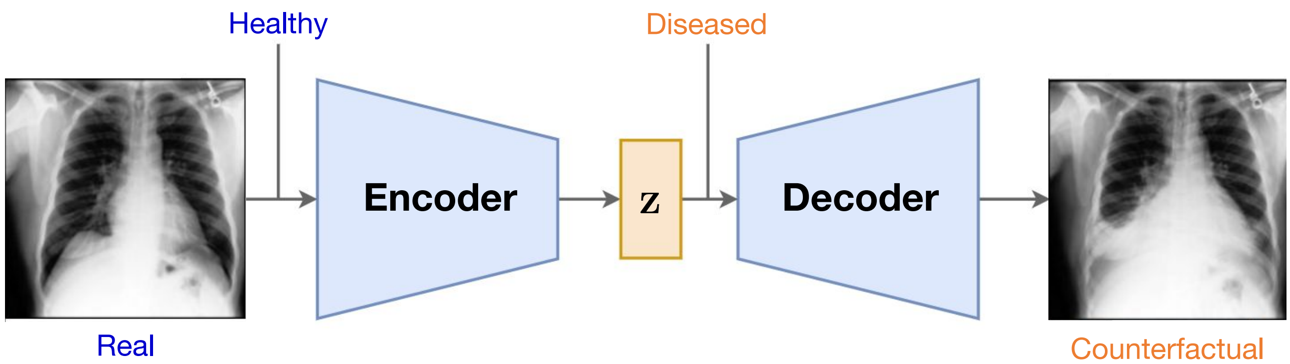
Research Questions:

- Can we generate plausible counterfactuals of medical images?
- How do we evaluate them?

Pawlowski, Castro, Glocker. Deep Structural Causal Models for Tractable Counterfactual Inference. *NeurIPS* 2020

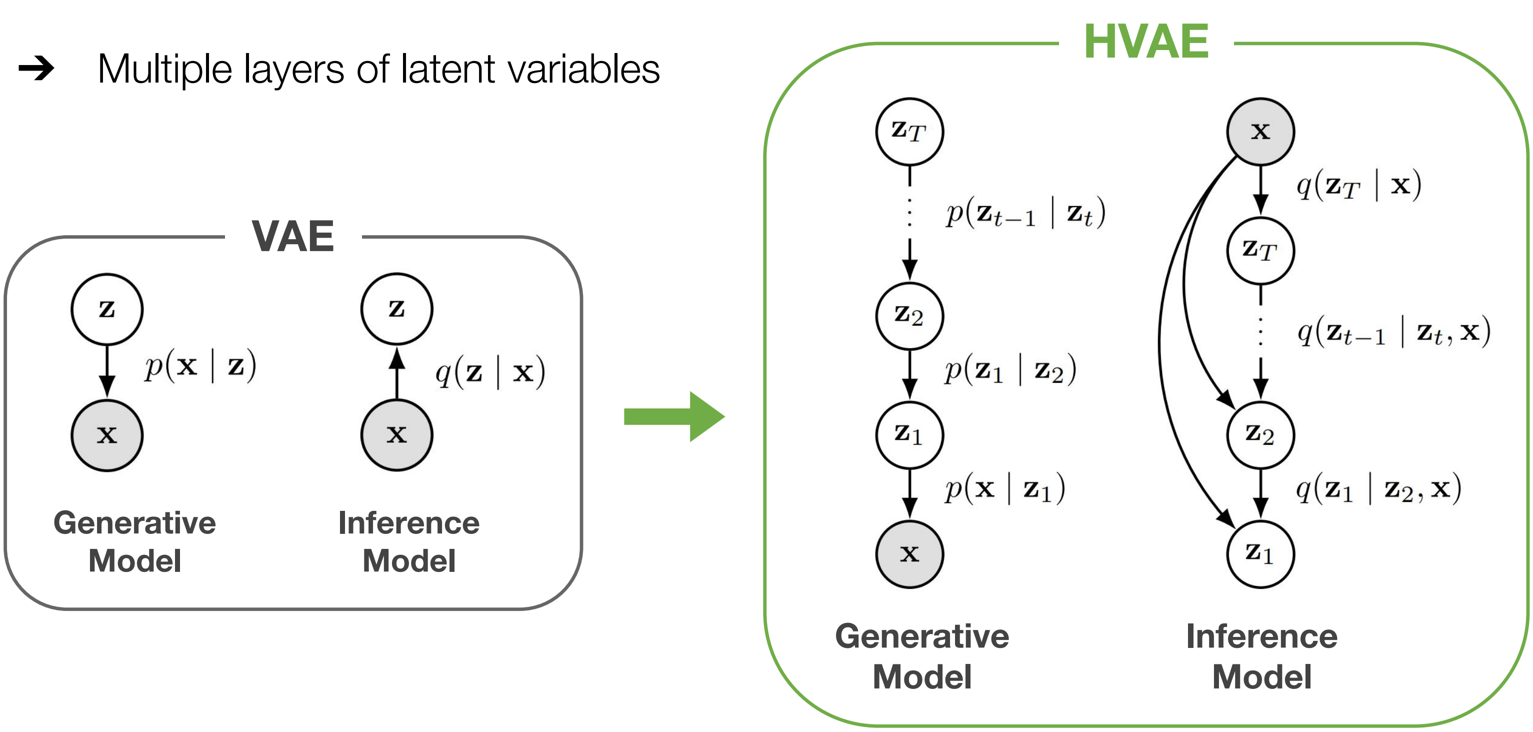
Causal Mechanism: Variational Autoencoder

➔ Example: computing an image counterfactual



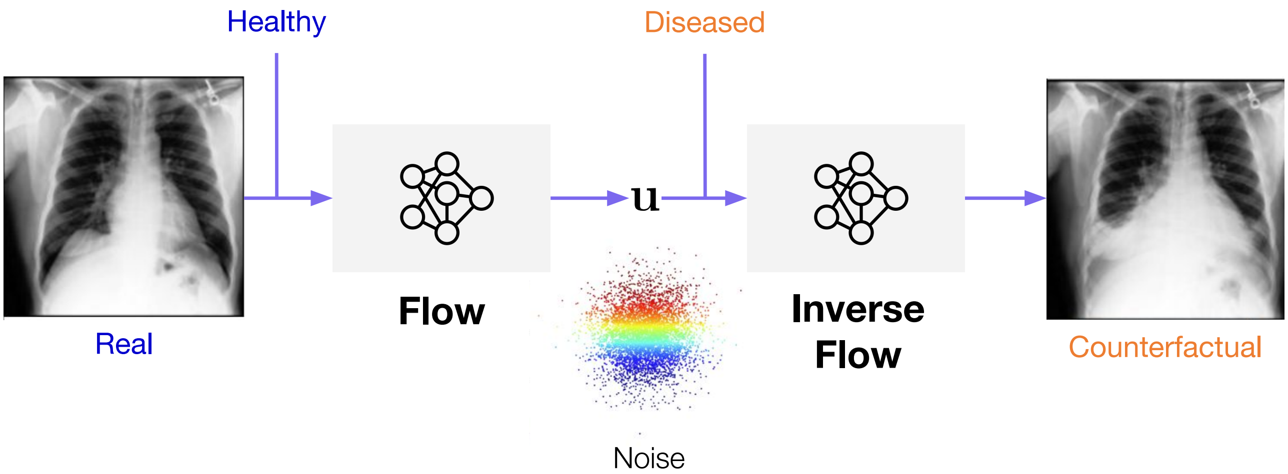
Causal Mechanism: Hierarchical VAE

➔ Multiple layers of latent variables



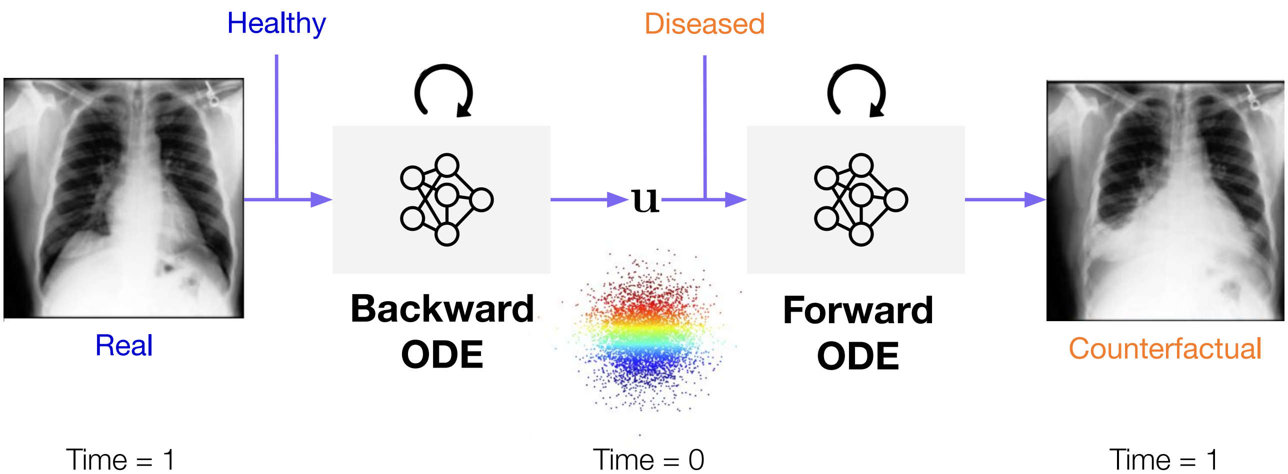
Causal Mechanism: Normalizing Flow

➔ Example: computing an image counterfactual



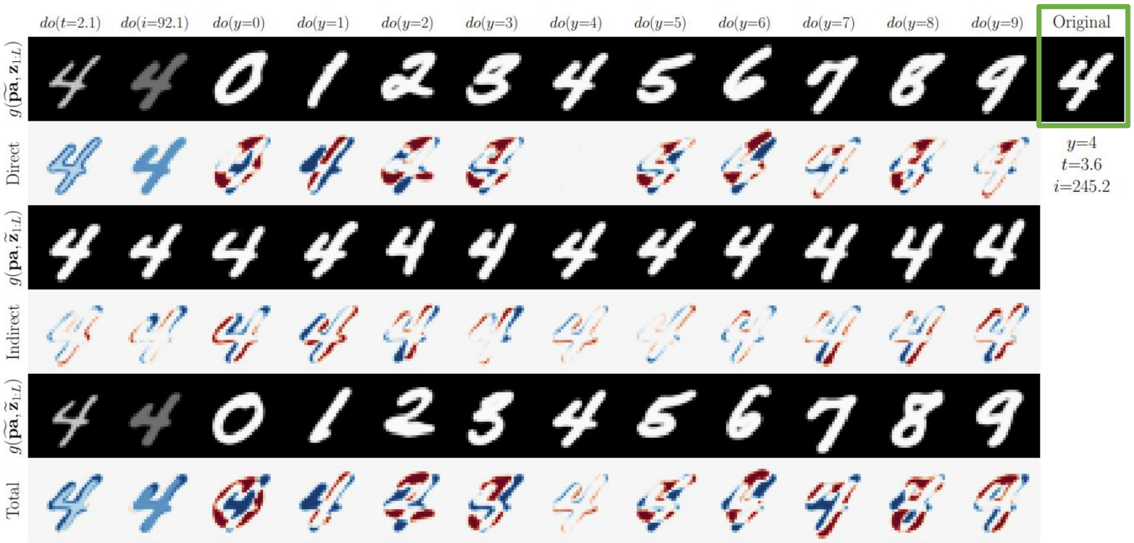
Causal Mechanism: Continuous Normalizing Flow

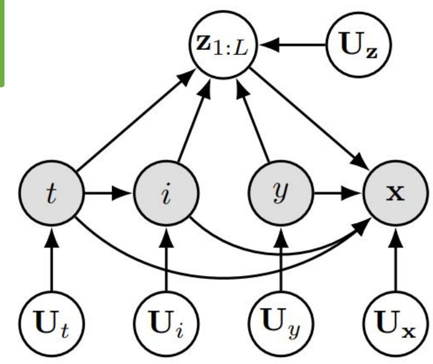
➔ Example: computing an image counterfactual



Case Study: Morpho-MNIST

Hugging Face
Code & Demo

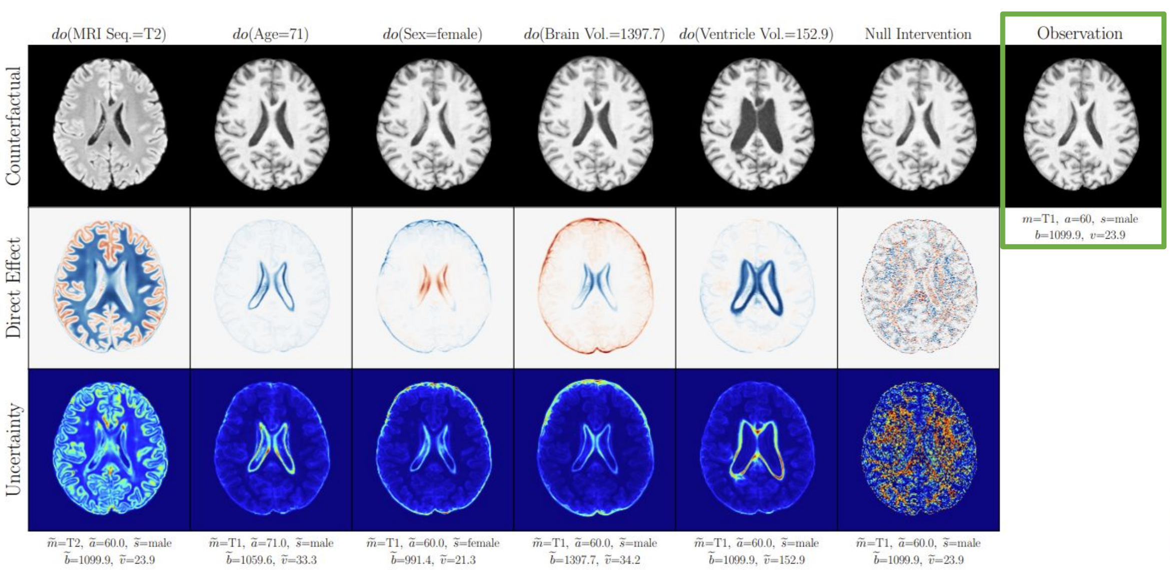


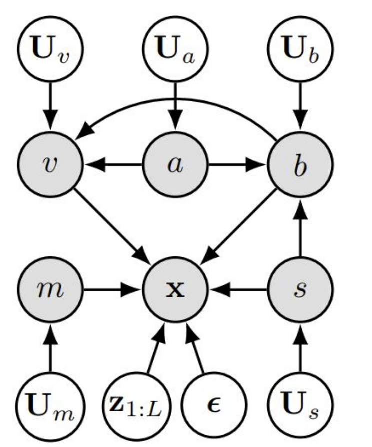


(a) Latent mediator SCM for Morpho-MNIST. Observed variables in the graph: image ($\mathbf{x}$), digit (y), stroke thickness (t) and pixel intensity (i).

Case Study: Brain Imaging

Hugging Face
Code & Demo



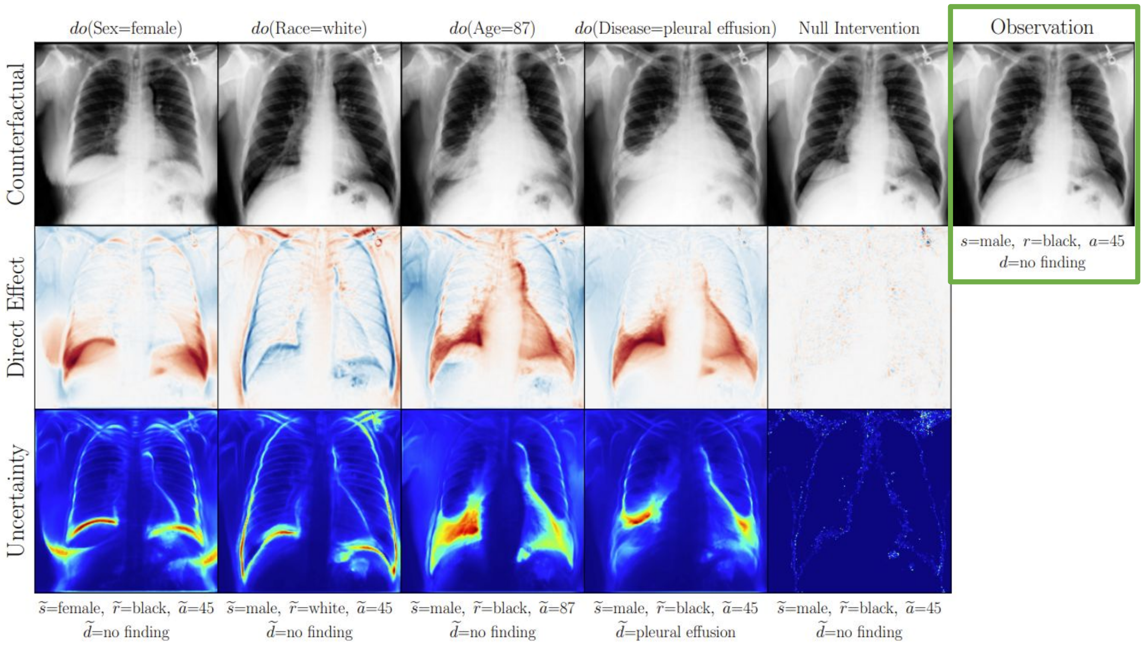


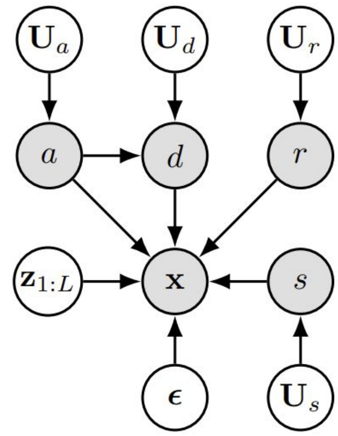
(a) Deep SCM for UK Biobank. MRI Seq. (m), age (a), sex (s), brain (b) & ventricle (v) volume.

De Sousa Ribeiro, Xia, Monteiro, Pawlowski, Glocker. High Fidelity Image Counterfactuals with Probabilistic Causal Models. *ICML* 2023

Case Study: Chest X-ray Imaging

Hugging Face
Code & Demo





(a) Deep SCM for MIMIC-CXR. The variables in the causal graph are: age (a), sex (s), race (r), disease (d) and chest x-ray $(\mathbf{x})$. The disease d is pleural effusion.

De Sousa Ribeiro, Xia, Monteiro, Pawlowski, Glocker. High Fidelity Image Counterfactuals with Probabilistic Causal Models. *ICML* 2023

Evaluating Counterfactuals: Axiomatic Properties

➔ There are **3** axiomatic properties of counterfactuals:

1. Composition   2. Effectiveness   3. Reversibility

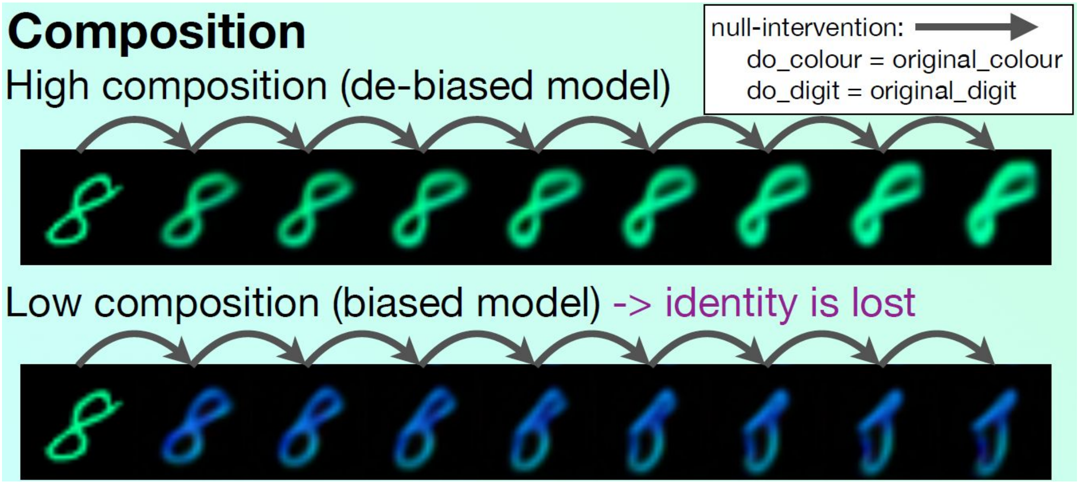
➔ They are necessary and sufficient in all causal models (Galles & Pearl, 1998; Halpern, 1998)

We can evaluate counterfactuals by **measuring** these properties!

David Galles and Judea Pearl. An axiomatic characterization of causal counterfactuals. Foundations of Science, 3:151–182, 1998.
Joseph Y. Halpern. Axiomatizing causal reasoning. In Proceedings of the Fourteenth Conference on Uncertainty in Artificial Intelligence, UAI'98, San Francisco, CA, USA, 1998.

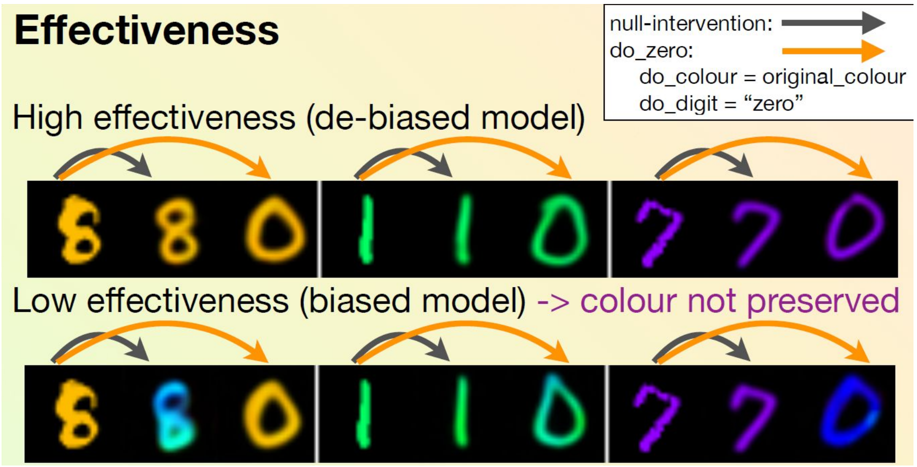
Axiom 1: Composition

Definition (informal)**.** Intervening on a variable to have the value it would otherwise have without the intervention will not affect other variables.



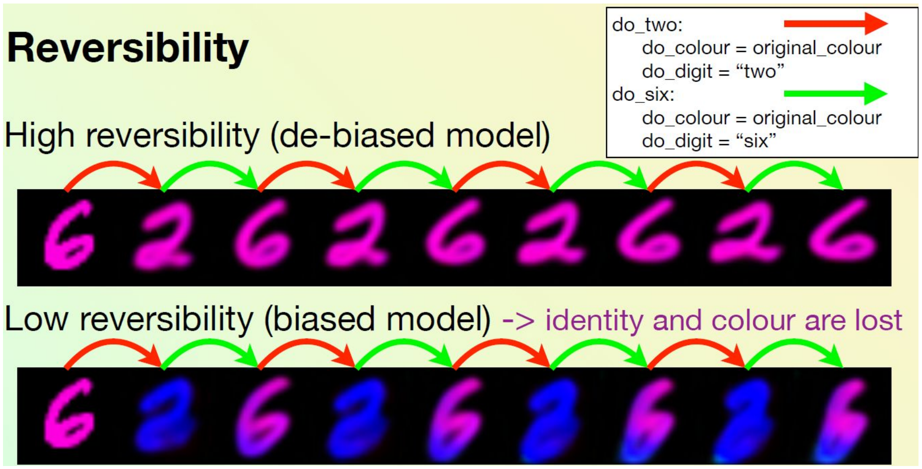
Axiom 2: Effectiveness

Definition (informal)**.** Intervening on a variable to have a specific value will cause the variable to take on that value.



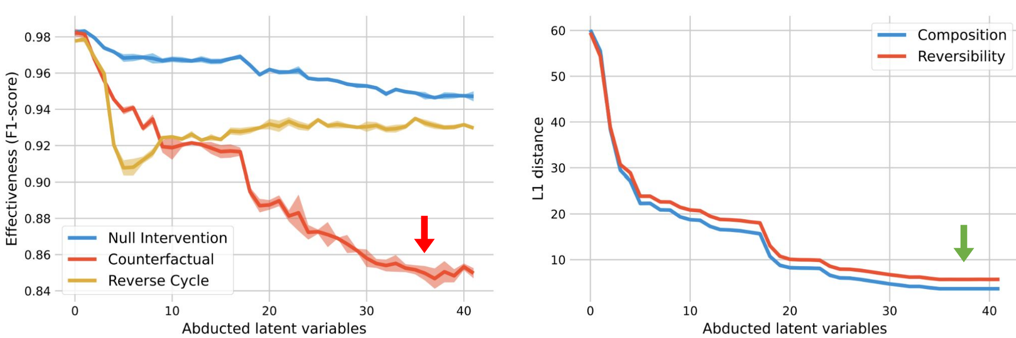
Axiom 3: Reversibility

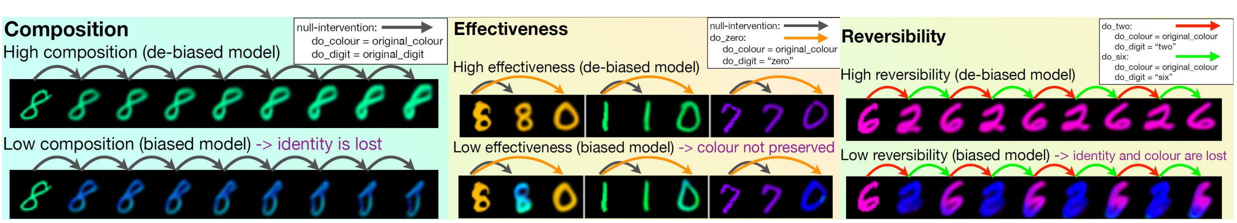
Definition (informal)**.** If setting a variable X to a value x results in a value y for a variable Y, and setting Y to y results in x for X, then X and Y will naturally take on the values x and y.



A Trade-off: Axiomatic Properties

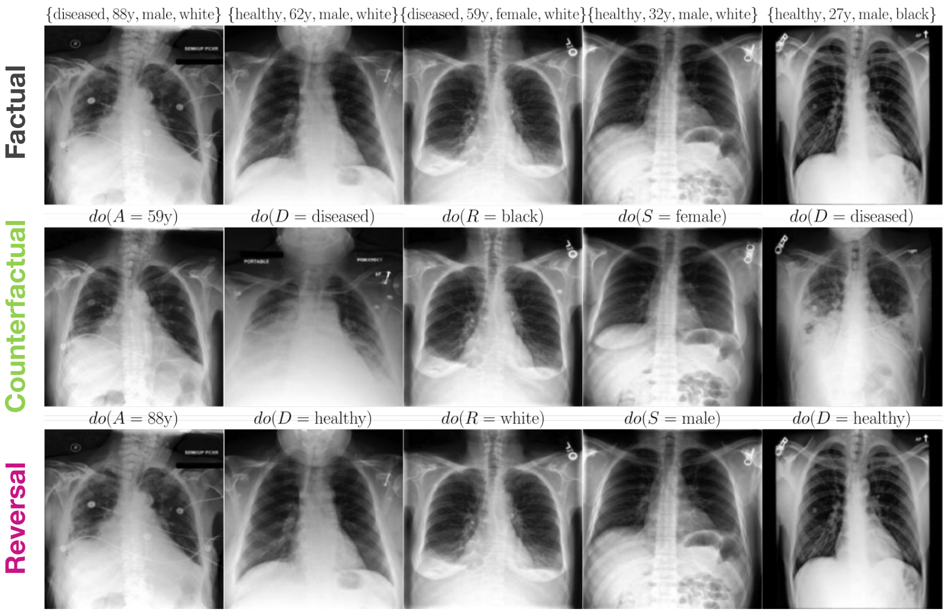
➔ **TL;DR:** We identify an inherent trade-off between the axioms

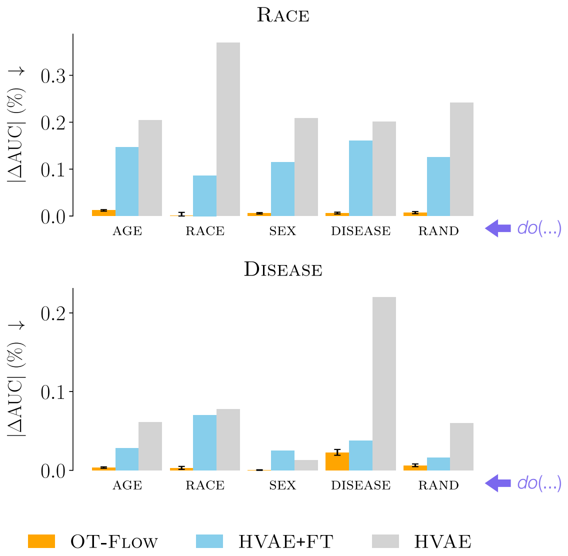




Monteiro*, De Sousa Ribeiro*, Pawlowski, Castro, Glocker. Measuring Axiomatic Soundness of Counterfactual Image Models. *ICLR* 2023

Revisit: Chest X-ray Counterfactuals





De Sousa Ribeiro, Santhirasekaram, Glocker. Counterfactual Identifiability via Dynamic Optimal Transport. *NeurIPS* 2025

Conclusion

➔ Generative AI holds great potential for healthcare applications

➔ With current technology, we can already generate plausible counterfactuals

➔ Thinking causally is crucial to identify and mitigate biased predictions

➔ Future AI systems should be aligned through causal world models to ensure safe and reliable decision-making

IMPERIAL

Hugging Face
Code & Demo

Thank you

Happy to take questions!


Fabio S. Ribeiro


Miguel Monteiro


Nick Pawlowski


Daniel C. Castro


Tian Xia


Ben Glocker

CHAI
BioMedIA
project mira ERC StG 2017
Microsoft